TakeSure — Confidence before taking something

Scan both. Check the pair. Ask the right questions.

An Ayurveda × Allopathy herb–drug interaction checker for Indian patients, doctors and the Ministry of Ayush

HACK@BVP 8.0 · Abstract Round

Team Bug Hunters

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DOMAIN: Health & Well-Being (AI4Impact secondary) · SDG 3 · SDG 9 · SDG 10

One patient's story — the gap today, and what TakeSure would do about it

1. **Mother** takes Thyronorm for thyroid and starts ashwagandha for stress.
2. **Nobody asks, nobody tells.** Her doctor never hears of it; the label says nothing.
3. **She photographs both.** TakeSure reads the names, confirms them and checks the pair.
4. **AMBER.** A cited source, timing advice, three questions. One tap sends the PDF to her doctor.

1 The problem

- **Two medical systems, zero cross-checking.** A thyroid patient on Thyronorm adds ashwagandha; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul.
- **Each pair can cause harm:** ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- **Nobody asks, nobody tells.** The doctor rarely asks about herbs, the patient rarely mentions them, the chemist has no tool, the label says nothing.
- **Existing tools do not cover it.** Medscape, Drugs.com and 1mg check only allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37% of hospital patients use allopathic and herbal medicines together

65% of herbal users do not know the dose, side effects or interactions

19% of Indian patients ever disclose herbal use to their doctor

1,473 ASU&H adverse-event reports received nationally since 2018

2 Proposed solution

- **Scan both.** The patient photographs the allopathic prescription and the Ayurvedic product label (or types the names); every name is confirmed before checking.
- **Match Indian names.** Brand → generic (Thyronorm → levothyroxine) and product → herbs (Chyawanprash → amla, ashwagandha…) with Sanskrit / Hindi / Latin synonyms.
- **Decide from data.** A rule engine checks every herb–drug pair against a curated, cited knowledge base: GREEN / AMBER / RED / NOT IN DATABASE, with timing advice.
- **Explain simply.** AI rewrites only the matched evidence into plain Hindi or English, with sources and three questions to ask the doctor.
- **Report in one tap.** A one-page PDF for the doctor and a pre-filled suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

> ***Our rule: The verdict comes from published data; AI only explains it.***

3 Approach — project workflow

Prescription photo allopathic medicines

Product photo Ayurvedic label / brand

› **1 · OCR** read both images; user confirms names

› **2 · Match** brand → generic; product → herbs; synonyms

› **3 · Knowledge base** cited herb–drug table; grade A/B/C

› ***Rule engine decides severity; AI does not***

› GREEN / AMBER / RED / NOT IN DB

› **4 · AI explains** simple Hindi/English + 3 doctor questions

› **Doctor PDF** verdict + sources

ADR report Ministry of Ayush format

Verdict rules: Worst row wins; an unknown pair is never shown as green; grade-C evidence caps at AMBER; the answer is identical with the AI switched off.

Feature	Medscape · Drugs.com · 1mg	MSKCC · Natural Medicines	TakeSure
Ayurveda × allopathy pair check	✗	~ single herbs	✓ every pair
Indian brands & Hindi	~	✗	✓ brands + synonyms
Source on every verdict	✓	✓	✓ + evidence grade
ADR report to Ministry of Ayush	✗	✗	✓ one tap

Our assessment from each tool's public documentation.

Stack. Next.js + TypeScript PWA (Hindi/English UI) · FastAPI + Pydantic REST API · PostgreSQL in production, SQLite offline · Tesseract OCR · RapidFuzz matching (≥85 accept, 70–85 ask the user) · Groq LLM, JSON-schema output, temperature 0 · ReportLab PDFs. No model training; runs fully offline on a laptop.

Herb × drug	Verdict	What happens	Source
Ashwagandha × Levothyroxine	AMBER · grade B	May raise thyroid hormone	Sharma 2018, PMID 28829155
Guggul × Warfarin	RED · grade B	May increase bleeding risk	MSKCC About Herbs
Giloy × Tacrolimus	RED · grade C	May oppose immuno-suppression	Nagral 2021, LiverTox

Knowledge base: 40–60 herb × drug pairs curated from PubMed, NIH LiverTox, MSKCC, WHO monographs and the Ayurvedic Pharmacopoeia, graded on the Stockley's model. Decision support only, never diagnosis.

4 Expected impact

Patients & caregivers A sourced answer before the first dose, in their own language, instead of silence or a guess. Free for citizens.

Ministry of Ayush Structured, reproducible ADR reports and real co-use data from the catchment of its 99 peripheral pharmacovigilance centres.

Doctors, vaidyas & chemists Finally learn what patients actually take; a QR on the chemist's bill or an API inside hospital and telemedicine systems makes the check routine.

Novelty Nothing public today checks an Ayurvedic product against an allopathic prescription. TakeSure builds the missing bridge table: Indian brands × Ayurvedic herb synonyms × graded evidence, open and auditable. Honest by design: grades, timing gaps instead of alarms, "not in database" instead of guesses.